

What actually works

FEATURES TRACED 41 UI → logic → data → render	PRODUCTION READY 33 tested end to end	PARTIAL 5 works, with a caveat	NOT READY 3 unverifiable or absent	DEFECTS FIXED 11 4 would lose data	BUTTONS WIRED 95/95 no dead controls
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Read this first. This environment blocks the Supabase host, so **no result here was obtained against the live database**. Persistence is proven against the real handlers and real pages with the database stubbed — which catches logic faults but not a wrong table name, a missing column, or a row-level-security rule. Two minutes on the deployed site closes that gap: open the designer, press Save, and confirm no red banner appears.

01 Feature inventory

Every feature a user can see or click, traced through its whole chain. A link marked **DB** means that link is required and missing; **DB** grey means not applicable to this feature.

FEATURE & LOCATION	CHAIN	TESTED	READY	NOTES / KNOWN BUGS
Customer intake form intake.html	FE BE DB CALC API FILE	YES	YES	Logo pointed at a file:/// path on one Mac — broken for every customer. Fixed.
Intake → AI sizing → draft quote api/intake-submit.js	FE BE DB CALC API FILE	YES	YES	Returned ok:true even when nothing saved. Now reports saved/notified and 502s when neither. 6 tests.
Customer record (create / edit) —	FE BE DB CALC API FILE	NO	NO	No customer table or screen exists. A customer is fields on a quote or design, re-typed each time.
Job creation (ServiceM8) api/create-job.js	FE BE DB CALC API FILE	NO	NO	Guards its key and returns a clear 500, but no SERVICEM8_API_KEY here — never executed against the real API.
Internal quote builder admin.html	FE BE DB CALC API FILE	YES	YES	Loads clean, no console errors. Design import block wired.
Quote tool (5-step) index.html	FE BE DB CALC API FILE	YES	YES	Was a blank page without a CDN — React now served from this origin with a real error state.
Quote save api/savequote.js	FE BE DB CALC API FILE	YES	YES	Correctly forwards the database's real status code rather than swallowing it.
Customer sign page sign.html	FE BE DB CALC API FILE	YES	YES	Renders a real quote; handles both no-id and load-failure with clear messages.
Price setup (brands / models) index.html	FE BE DB CALC API FILE	YES	YES	Showed a green tick regardless of whether prices saved. Now withholds it and warns.
Floor plan upload — image designer.html	FE BE DB CALC API FILE	YES	YES	JPG/PNG, downscaled and tiled for reading.

FEATURE & LOCATION	CHAIN	TESTED	READY	NOTES / KNOWN BUGS
Floor plan upload — PDF designer/ui/pdf.mjs	FE BE DB CALC API FILE	YES	YES	pdf.js vendored same-origin; multi-page with page switching.
AI plan reading (tiled) api/plan-read.js	FE BE DB CALC API FILE	YES	YES	Overlapping full-res tiles. Key missing → clear error, and no rooms invented.
Numbers read — review & correct Plan tab	FE BE DB CALC API FILE	YES	YES	Every number listed and editable; an edit rebuilds chains and re-measures rooms.
Dimension-chain reconstruction engines/chains.mjs	FE BE DB CALC API FILE	YES	YES	350-2050-220-3400-90-2720 → 0-350-2400-2620-6020-6110- Deterministic code, not prompt text.
Manual scale calibration engines/calibration.mjs	FE BE DB CALC API FILE	YES	YES	600 px / 6000 mm = 10 mm/px; 320 px = 3200 mm. Two-point picking fixed earlier this session.
Room detection from labels designer/app.mjs	FE BE DB CALC API FILE	YES	YES	Chain pixel-anchoring fixed — rooms were landing in the wrong part of the house.
Printed room sizes (brochure plans) engines/dimensions.mjs	FE BE DB CALC API FILE	YES	YES	“4.3 x 7.1m” read at 98% confidence, no calibration needed.
Room verification & editing Rooms tab	FE BE DB CALC API FILE	YES	YES	An edit propagates: 10 kW → 14 kW, airflow 500 → 700 L/s, outlets 5 → 7, price 13,500 → 14,999.
HVAC load calculation engines/loads.mjs	FE BE DB CALC API FILE	YES	YES	3 m × 4 m = 12 m² reaches the engine as exactly 12. 145 W/m² rule preserved and cross-checked.
Equipment selection engines/equipment.mjs	FE BE DB CALC API FILE	YES	YES	Was aiming 10% high and buying a model size too big on every job. Now targets the design load.
Manufacturer specs (489 models) engines/unit-specs.mjs	FE BE DB CALC API FILE	YES	YES	Data existed but was never imported — airflow and static checks were skipped on every job. Now wired.
Zone controller selection engines/catalogue.mjs	FE BE DB CALC API FILE	YES	YES	Costed from the MMEM quotation; brand-locked where the controller is.
Airflow calculation engines/airflow.mjs	FE BE DB CALC API FILE	YES	YES	50 L/s per kW, per-room share, minimum-airflow floor.
Outlet design engines/outlets.mjs	FE BE DB CALC API FILE	YES	YES	Type and count from airflow and throw; split rule for long rooms.
Duct design engines/ducts.mjs	FE BE DB CALC API FILE	YES	YES	Velocity = Q/A verified by hand on all 8 stocked diameters. 450/500 removed — not installed.
Return air engines/returnair.mjs	FE BE DB CALC API FILE	YES	YES	Takes the unit's own return connection from the manufacturer sheet: 1×400 or 2×350/400.
Zoning engines/zones.mjs	FE BE DB CALC API FILE	YES	YES	Open-plan grouping, minimum-open airflow, no bypass assumed.

FEATURE & LOCATION	CHAIN	TESTED	READY	NOTES / KNOWN BUGS
Static pressure engines/pressure.mjs	FE BE DB CALC API FILE	PART	PARTIAL	Now has real available-static figures for 110 of 489 models; the rest still declare SPECIFICATION DATA REQUIRED.
Bill of materials engines/bom.mjs	FE BE DB CALC API FILE	YES	YES	32 lines on the sample job. Buys whole 6 m lengths, not metres.
Supplier pricing (MMEM) engines/supplier-pricing.mjs	FE BE DB CALC API FILE	YES	YES	62 ducted sets, 20 controls, 32 accessories, 5 paircoil sizes — dated and attributed.
Material rates settings Settings → Materials	FE BE DB CALC API FILE	YES	YES	Shows the part code and pack price behind each rate, so real prices and placeholders are distinguishable.
Commercials, margins, totals engines/costing.mjs	FE BE DB CALC API FILE	YES	YES	16 tests against hand-computed values. GP equals the fee exactly at any cost.
Design → quote transfer engines/store.mjs	FE BE DB CALC API FILE	YES	YES	Crashed on any design missing a stage — estimator got an error, no quote. Fixed. Pressing twice does not duplicate.
Design save / load engines/store.mjs	FE BE DB CALC API FILE	YES	YES	Reported “Design saved” when it had only reached the iPad. Now warns and keeps the Save button live.
Revisions & restore engines/model.mjs	FE BE DB CALC API FILE	PART	PARTIAL	Works, but the picker is a native prompt() listing revisions — awkward on an iPad.
Reports (internal + customer) designer/ui/reports.mjs	FE BE DB CALC API FILE	YES	YES	Verified generating a real Internal HVAC Design Sheet. Opens a print window — no PDF library; you print to PDF.
NAC Design Assistant (AI) api/design-assistant.js	FE BE DB CALC API FILE	PART	PARTIAL	Guards its key and answers from the design only. Not exercised against the live Anthropic API from here.
HVAC design settings Settings screen	FE BE DB CALC API FILE	YES	YES	Every assumption editable and persisted; reset-to-defaults per section.
Warnings & acknowledgement engines/warnings.mjs	FE BE DB CALC API FILE	YES	YES	Severity-ranked, acknowledgeable, and blocking where it should block.
Navigation & tabs designer.html	FE BE DB CALC API FILE	YES	YES	13 tabs, 8 step markers, all wired. No horizontal overflow at iPad width.
iPad / touch behaviour all pages	FE BE DB CALC API FILE	PART	PARTIAL	1080×810 landscape: no overflow, touch calibration works. Native prompt()/confirm() dialogs are the weak spot.
Authentication —	FE BE DB CALC API FILE	NO	NO	There is none. Anyone with the URL can open admin.html and the designer, and the database key ships in the page.

02 Defects found and fixed

Found by running the application, not by reading it. The first four were silently losing data — nothing on screen revealed them.

CRITICAL The intake form thanked customers for submissions nobody received

`api/intake-submit.js` returned `ok:true` whatever happened. A missing key skipped the write, a thrown error was swallowed as “non-fatal”, and `fetch` resolves on a 401 — so a *rejected* write looked identical to a successful one.

CRITICAL “Design saved” when it had only reached the iPad

`setSetting` returned true regardless of the database's answer, so the designer cleared its dirty flag and the work existed on one device only. The warning is now raised inside `storage.set` itself, where no caller can forget it.

CRITICAL Pushing a design to a quote crashed on any design missing a stage

`designNotesBlock` read `s.legacy.kw` unguarded. The estimator got an exception and no quote. Every line of that block is now optional.

CRITICAL The quote tool was a blank white page without a CDN

React loaded from unpkg.com with no fallback and no error state. Now served from this origin, CDN as fallback, and a failure says which half is missing.

HIGH Every job was quoted on a unit one size too big

Equipment selection took its target as the midpoint of the acceptance window — 1.10× the design load — on top of the 5% safety margin already in the load. An 18 kW unit outranked a 16 kW on a 15.71 kW job.

HIGH 489 models of manufacturer data sat unused

`catalogue.mjs` never imported `unit-specs.mjs`, so airflow and static-pressure checks were skipped on every job for want of data the app already had.

HIGH A room with one dimension silently counted as zero

It stayed on the schedule, contributed nothing to the load, and raised no warning — the system just came out a size small. Now blocked as “Needs a dimension”.

HIGH Room labels landed in the wrong part of the house

Chain stations are millimetres from the chain's own zero; the code converted pixels from the *image's* origin. A label at 4,500 mm resolved to 12,500 mm.

HIGH Brochure plans read as nothing at all

The tool only understood chained dimension strings. Plans that print “LIVING 4.3 x 7.1m” inside each room — most of what an estimator is sent — produced no rooms, despite the highest-confidence code path already existing unused.

HIGH The customer intake logo pointed at a path on one Mac

`file:///Users/nacelectrical/Downloads/...` — a broken image for every customer who opened the form.

HIGH

W.I.R. was being air conditioned

The exclusion rule matched “WIR” but not “W.I.R.”, which is how half of Australian plans write it. Same for W.C. and B.I.R.

03 Remaining risks

Nothing here is broken. These are the things I could not verify from this environment, or that are working as built but carry a risk worth your decision.

No authentication, anywhere

admin.html, designer.html and the price setup screen are open to anyone with the URL, and the Supabase anon key ships in the page source. Whether that matters depends on how you share those links — but it is a decision to make deliberately, not one to discover.

The live database is unverified from here

Every persistence result was obtained with the database stubbed. Logic faults are covered; a wrong table name, a missing column or a row-level-security rule would not be. Press Save on the deployed designer and confirm no red banner.

No customer or job records

Your brief lists customer creation, editing and job creation as features. They do not exist — a customer is fields on a quote or a design, re-typed each time. I have not invented tables to make the inventory look complete.

ServiceM8 job creation never executed

api/create-job.js guards its key and returns a clear error, but with no SERVICEM8_API_KEY in this environment it has never run against the real API.

Native prompt() dialogs on iPad

Design list, revisions and report choice all use prompt(). They work, but they are awkward on a tablet and some browsers suppress them. Worth replacing with a proper picker.

Reports print rather than export

Report generation opens a formatted print window. There is no PDF library — you print to PDF. That is a real workflow, but it is not a download button.

379 of 489 models still lack a static-pressure figure

The tech sheets do not carry it for every model. Those declare SPECIFICATION DATA REQUIRED rather than guessing, which is correct — but the static check is skipped for them.

Placeholder material rates still in play

13 BOM lines on a typical job use shipped placeholder rates, not NAC's. On job-cost-plus-fee those go straight to the customer. Isolator, cable, hanging strap, drain tray and mounting feet are the ones to price.

Verified by execution: 283 unit tests · 8 Playwright suites in tools/browser-tests/

Full workflow walked in one run – plan → 16 rooms → 18.63 kW → equipment → 32-line BOM → \$17,016.54 inc GST → save → refresh → quote → sign page.